

RadioFrequency Bridge for Pulsed Nuclear Magnetic Resonance

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Radio-Frequency Bridge for Pulsed Nuclear Magnetic Resonance*

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A symmetrical rf bridge for pulsed nuclear magnetic resonance experiments is described. The bridge is non-microphonic and permits the quick damping of the transients from rf driving pulses without the degrading of free induction decay signals.

I. INTRODUCTION

IN pulsed nuclear magnetic resonance experiments on samples that have short free induction decay times, one tries to use rf pulses whose width t_w is much less than the decay time. To nutate the macroscopic magnetization through a large angle in time t_w , the oscillating magnetic field producing the nutation must be large, and this requires a large amount of rf power (about a kilowatt) to be supplied to the coil producing the oscillating magnetic field. After the rf pulse ends, the precessing macroscopic magnetization induces a voltage (millivolts or less) in the same "driving coil" or a second "pickup coil" placed at right angles to the driving coil. The signal is then amplified, detected, and observed.

At the end of the rf pulse the driving coil normally has a kilovolt or more of excitation voltage across it. This excitation voltage decays exponentially with a time constant $T=Q/(\pi f)$, where Q is the Q of the driving tank and f is its resonant frequency. One has difficulty observing the picked-up signal voltage during the time that the transient excitation voltage is much greater than the picked-up signal voltage. This can be a serious problem if the time it takes the excitation voltage to decay to acceptable levels is as long as or longer than the free induction decay times, a situation that often arises in solids. In a bridge, one cannot lower the Q of the driving tank to an arbitrarily small value to decrease the transient decay time because the size of the picked-up signal decreases with decreasing Q .

The signal induced in the pickup coil is very small and must be processed by a high-gain broad-band amplifier that is closely coupled to the pickup coil before the signal is observed. The rf pulse must be prevented from reaching this amplifier to protect the amplifier from damage and keep it from "blocking." In most pulsed NMR apparatus this is done by using devices developed for cw nuclear magnetic resonance experiments. Examples of these are the

crossed-coil system,¹ the symmetrical bridge,² and the asymmetrical bridge.³ All these devices are quite microphonic and have the following further disadvantages: the crossed-coil system is not well suited to experiments involving temperature regulation; no two adjustments can be used to approach "balance" in an orthogonal manner for the symmetrical bridge; the asymmetrical bridge can be balanced at only one frequency and is ineffective when used with short rf pulses that have a broad frequency spectrum.

In the next section a symmetrical bridge is described which does not depend critically upon good balance for preventing the rf driving pulse from reaching the receiver and is thus not microphonic. It further can be operated so that the sample coil has a low Q while the rf pulse is applied and its transient is dying out, and a high Q while the free induction decay signal is being observed.

II. rf BRIDGE

As is well known, semiconductor diodes have a sharp corner in their current vs voltage characteristic. We take advantage of this characteristic in the rf bridge shown in Fig. 1. Coil L_1 and capacitor C_1 form the resonant circuit

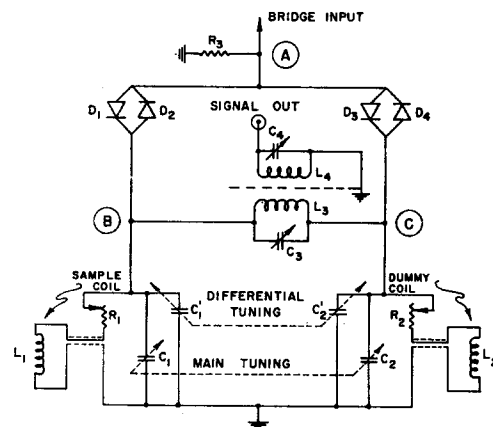


FIG. 1. Diode bridge. D_1 , D_2 , D_3 , and D_4 are FD-100 diodes.⁴ R_1 and R_2 are 25- Ω composition potentiometers. $R_3=1000\ \Omega$.

¹ F. Bloch, W. W. Hansen, and M. Packard, Phys. Rev. **70**, 474 (1946).

² H. C. Torrey, Phys. Rev. **75**, 1059 (1949).

³ H. L. Anderson, Phys. Rev. **76**, 1460 (1949).

⁴ Fairchild Semiconductor Corporation, San Rafael, California.

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† Alfred P. Sloan Fellow.

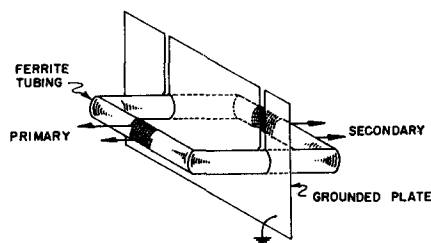


FIG. 2. Output transformer with Faraday shield.

containing the sample. Coil L_2 and capacitor C_2 are a dummy resonant circuit or tank matched as closely as possible to L_1 and C_1 . By virtue of their being high- Q parallel resonant circuits, the two resonant circuits have high impedances. L_3 and C_3 are the primary coil and tuning capacitor of the output transformer. Diodes D_1 and D_2 couple the sample tank to the driving radio-frequency source, while diodes D_3 and D_4 couple the dummy tank to the rf source.

When a large rf voltage is applied to the bridge at point A, the sample tank and dummy tank are excited. Since the impedances of these tanks are much larger than that of the diodes, only a small voltage drop appears across the diodes. If the diodes and tanks are perfectly matched, no voltage difference appears across the primary of the output transformer (points B and C). Even if the tanks or diodes are not precisely matched, only a small voltage appears across the primary of the output transformer because of the comparatively low diode impedance. Since close matching of the tanks is not critical, temperature variations and microphonics are not a problem.

The Q of the tanks is very low while the voltage on the tanks is over approximately 1 V because point A is connected to ground through the 1-k Ω resistor R_3 . The driving pulse is thus quickly damped for voltages above this level.

When the voltage across the tanks is less than a few tenths of a volt, the diodes revert to their high-impedance state and decouple points B and C from the input as well as from each other. Any signal voltage that develops across the sample coil now appears across the primary of the output transformer and the dummy tank.

Diodes D_1 , D_2 , D_3 , and D_4 can adversely affect the signal voltage in at least four different ways. The first is by the diodes taking an excessively long time to switch from the low-impedance to the high-impedance state. The second is by the diodes generating enough noise that it appreciably adds to the noise from the rest of the apparatus. The third is by the diodes having a low enough resistive impedance in the high-impedance state that the diodes lower the Q 's of the various resonant tanks of the bridge. The fourth is by the diodes having a large enough capacitance in their high-impedance state that resistor R_3 damps the sample

tank and dummy tank. The FD-100 diode,⁴ used in the bridge described in Sec. IV, has sufficiently satisfactory characteristics that it does not degrade the signal voltage in any observable manner. (There are undoubtedly many other diodes with equally or more satisfactory characteristics. No exhaustive search or study was made to find them.) Direct observation showed that its switching speed under the conditions in which it is used is much less than a microsecond, and thus completely satisfactory. The tests for the second and third characteristics were made by placing diodes across the receiver coil of a crossed-coil rf head without observing any adverse effects. The test for the fourth effect was made by removing resistor R_3 without observing any improvement in the observed signal.

While the rf driving voltage is on, points B and C are at a very high voltage. Even if there is no voltage difference between them, a large signal may be coupled to the secondary by the capacitance between the primary and secondary coils. Therefore a grounded Faraday shield is used to reduce this capacitance. A convenient method of doing this is to wind the primary and secondary coils on a ferrite loop as shown in Fig. 2. This allows close magnetic coupling between the primary and secondary coils while preventing close capacitive coupling of the coils.

III. GATED DAMPING CIRCUIT

As previously mentioned, the resistor R_3 effectively damps the sample and dummy tanks only when the voltage on these tanks is greater than $\frac{1}{2}$ V. Below $\frac{1}{2}$ V the ringing of the tanks damps very slowly because of their high Q . The damping rate of these tanks can be considerably quickened below the $\frac{1}{2}$ -V level if one can keep the diode impedance independent of the voltage across the tanks. This can be accomplished by use of the circuit shown in Fig. 3. Tube V_1 is normally cut off. In order to damp the tanks, a positive pulse is applied to the grid of tube V_1 . Direct current then flows through diodes D_2 , D_4 , and D_6 , lowering their impedances. This allows the tanks to be damped by R_3 as well as the plate impedance of tube V_1 . The rf driv-

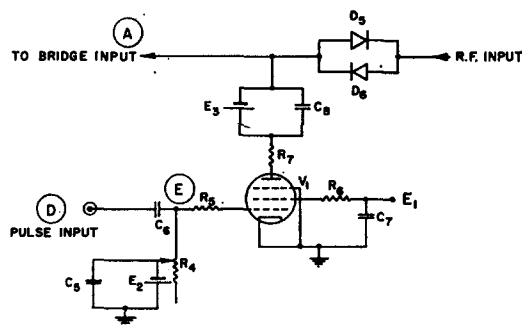


FIG. 3. Gated damping circuit. V_1 =6688 vacuum tube; E_1 =200 V; E_2 =10 V; E_3 =6.75 V; R_4 =1-k Ω potentiometer; R_5 , R_7 =47 Ω ; R_6 =100 Ω ; C_5 , C_7 , C_8 =0.1 μ F; D_5 , D_6 =FD-100 diodes.⁴

ing pulse is applied through diodes D_5 and D_6 to prevent tube V_1 from being shorted to ground at the bridge input A.

One must turn off the direct current through the tanks and diodes smoothly so that the tanks do not ring. This is done by use of a simple RC wave-shaping circuit which transforms the square wave input at D into an exponentially decreasing signal at the grid of tube V_1 . The rate of decrease of the damping current is adjustable by means of the potentiometer R_4 in the wave-shaping circuit.

IV. PERFORMANCE CHARACTERISTICS OF THE 10-Mc BRIDGE

A 10-Mc bridge based upon the circuits of Figs. 1 and 3 has performed satisfactorily for us for the past two years. As indicated in Fig. 1, we find it convenient to gang together the main tuning capacitors C_1 and C_2 such that their capacitances increase simultaneously. The small differential tuning capacitors C_1' and C_2' are ganged such that the capacitance of one increases while that of the other decreases. Small variable resistors R_1 and R_2 are placed in series with coils L_1 and L_2 to help match their Q 's. They also permit one to change easily from a high Q tank to one of low Q to increase the bandwidth of the circuit for samples with very short decay times.

The output transformer is constructed of type 4D ferrite tubing⁵ cut to form a square loop 5 cm on a side (see Fig. 2). The primary L_3 consists of 16 turns while the secondary L_4 has 4 turns. A transformation from 100 Ω to about 15 k Ω is obtained. The Faraday shield is simply a grounded metal plate placed between the coils.

For the damping circuit, two 6688 tubes are used in parallel rather than the single tube shown in Fig. 2. Their combined impedance to ground is approximately 500 Ω when gated on. A 20-V pulse from a Tektronix 163 pulse generator is applied at point D to turn the gating circuit on. The occurrence in time of the damping pulse can be varied by triggering the pulse generator from the sawtooth waveform produced by a Tektronix 162 waveform generator. On our apparatus a 10-Mc filter is placed before pulse input D to prevent leakage of rf energy to the sample through this point.

The sample and dummy coils used have an inductance of 2.5 μ H, and the resonant tanks have Q 's of approximately 65. Therefore the impedance of each tank is about 10 k Ω . With these coils, a 90° nutation for a sample containing protons is produced by a 750-V peak rf driving pulse 3 μ sec in length. The rise and fall times of the pulse are about 0.4 μ sec. A signal of 0.20 V peak is obtained from a water sample filling the sample coil, using an amplifier with a gain of 3000 and an input and output impedance of 90 Ω .

When the rf pulse is turned off, the damping circuit is

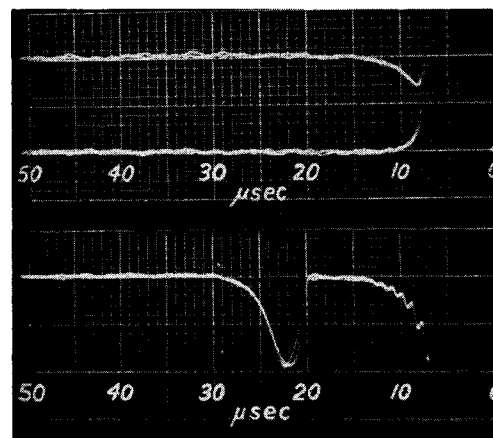


FIG. 4 (a) Top oscilloscope trace is the detected induction signal from protons in a water sample in a relatively homogeneous magnetic field. The rf excitation pulse lasts 3 μ sec, after which time the bridge is damped by a damping pulse. The lower trace is a base-line signal that is generated by applying a second rf pulse before the proton spin system has had time to develop an appreciable magnetization along the applied static magnetic field. (b) Same as top trace in (a) except that damping pulse for the bridge is applied at the 20- μ sec point instead of the 3- μ sec point.

gated on. The time constant of the RC circuit is set so that an exponentially decreasing voltage with a time constant of approximately 2 μ sec appears at point E. The rf voltage on the tank is then damped to 0.1 V at 1.5 μ sec after the end of the driving pulse. Without the damping current the tank voltages would be of this magnitude 6 μ sec after the pulse. It should be noted that this "undamped" case still benefits from the damping by the R_3 loading resistor which is effective at levels above 1 V. This should be compared to the situation in a "conventional" undamped bridge. Assuming the same Q and driving rf level, the 0.1-V level would not be reached until some 18 μ sec after the end of the rf pulse. (Of course loading to ground by 1 k Ω in series with a pair of diodes could also be used in a "conventional" bridge.)

Although the damping pulse reduces the impedance of the diodes to 100 Ω at the end of the rf pulse, their impedances are on the order of 50 k Ω by 2 μ sec after the end of the rf pulse. At this point, the signal from the sample begins to develop across the sample coil and output transformer with a time constant determined by the Q of the resonant circuits. For our apparatus, where the Q 's of the sample and dummy tanks are 65, the time constant is about 2 μ sec and the signal grows to 95% of its maximum value at 8.5 μ sec after the end of the rf pulse. The above described behavior of a signal from the bridge is shown in the upper trace of Fig. 4(a). This trace is the detected induction signal from the protons of a water sample in a magnetic field that is homogeneous enough that the transverse magnetization does not observably decay in 50 μ sec. The oscilloscope trace in Fig. 4(b) is taken under the same conditions as for the top trace in Fig. 4(a) except that the pulse that damps the bridge occurs 20 μ sec from the origin

⁵ Ferroxcube Corporation of America, Saugerties, New York.

instead of occurring 3 μsec from the origin. The transient from the rf pulse is now undamped when it falls below $\frac{1}{2}$ V, and the ringing of the tanks is displayed on top of the induction signal. The rapid decay of the rf signal voltage across the tanks on application of the damping pulse at 20 μsec is now displayed (limited by the bandwidth of the amplifier), along with the rf signal voltages subsequently built up on removal of the damping pulse. If the Q of the resonant tanks is reduced by a factor of 2, the signal reaches 95% of its peak value at 5.7 μsec after the end of the rf pulse.

Our procedure for balancing the bridge is as follows. We tune the sample tank, dummy tank, and transformer with a grid-dip meter and then obtain a free induction decay signal for a water sample. Using an rf pulse that produces

much less than a 90° nutation, we then maximize our free induction decay signal using the main tuning condensers. Without applying damping pulses, the differential condensers and resistors R_1 and R_2 are then used to balance the bridge to give a minimum blocking time for the rf amplifier. (One may have to readjust the main tuning condenser for maximum signal during this process.) Setting the pulse width for a 90° nutation, the damping pulse is turned on and adjusted to further reduce the blocking time. Once set, the bridge will remain balanced for days without any adjustment. When the temperature of the coils is varied between 77°K and room temperature, we do not need to rebalance the bridge, but just change the main tuning condenser to compensate for the change in inductance of the sample and dummy coils.

100-Mc Digitron—A Digital Time Analyzer*

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An instrument is described which measures microsecond time intervals by counting pulses from a 100-Mc CW oscillator; a commercially available scaler is employed as the major component. The over-all differential non-linearity of the instrument is $\leq 10^{-2}$. Data are stored in a magnetic core memory. The deadtime associated with measurement and storage has been minimized to permit use in experiments with pulsed accelerators.

I. INTRODUCTION

RECENT interest in the decay rate of positive muons and the time variation of the decay rate of negative muons bound in certain mu-mesic atoms has stimulated development of various digital instruments, "digitrons," for the precise measurement of time intervals in the microsecond range. All of these instruments¹⁻⁴ have several features in common: (a) Timing is done with reference to an oscillator of approximately 10 Mc. (b) The time range covered is usually 10 to 20 μsec divided into 0.1- μsec

intervals. (c) The information obtained is stored in the memory of a pulse-height analyzer in a "serial address" fashion resulting in instrument deadtimes of up to several hundred microseconds. The introduction of commercial 100-Mc decade scalers⁵ has made it practical to design and build an instrument with a finer time scale (a timing frequency of 100 Mc) and considerably less deadtime. The operating principles and some circuit details are presented below.

II. PRINCIPLE OF OPERATION

The fundamental operations performed in measuring and recording the time interval between a start pulse and a stop pulse are depicted in Fig. 1.

A start pulse opens a gate which allows a 100-Mc pulse train to be counted by the scaler until a stop pulse closes the gate. The number in the scaler, representing the time interval, is then recorded in the core memory and the scaler reset to zero. A practical form of the above apparatus must incorporate circuits to prevent interference from additional input pulses during the measurement and storage process,

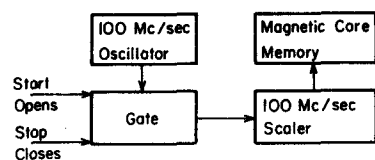


FIG. 1. Operating logic.

* Research supported by a joint program of the Office of Naval Research and the U. S. Atomic Energy Commission.

† Present address: Argonne National Laboratory, Argonne, Illinois.

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⁴ R. A. Lundy, Phys. Rev. 125, 1686 (1962).

⁵ Model 1060 manufactured by Eldorado Electronics Company, Berkeley 10, California.